The Topology of Reliable Cognition: Grounding Structure, Cognitive Distance, and the Dynamics of Abductive Overwrite

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April 2026

Abstract

We propose a structural framework linking symbol grounding and abductive dynamics in cognitive systems. Rather than treating grounding as a binary property, we model it as inducing a topology over a shared concept set, and introduce *cognitive distance* (d_{cog}) to quantify structural similarity between agents.

We operationalize d_{cog} via representational similarity analysis (RSA), defining a second-order correlation metric (ρ_{cog}) over concept-level activation patterns. Experiments on language models (Pythia-70m training checkpoints and GPT-2) show that relational structure becomes increasingly aligned through training, supporting the view that conceptual structure evolves continuously rather than via abrupt overwrites. Cross-model replication ($\rho_{\text{cog}} \approx 0.42$ between GPT-2 and fully trained Pythia vs. ≈ 0.10 vs. random initialization) suggests that training induces a shared representational geometry independent of architecture.

We further introduce *abductive overwrite* as a mechanism that restructures interpretive frames under hypothesis revision, and identify *negative abductive capability* (NAC) as a structural condition for maintaining grounding stability. Evidence from comparative cognition (*C. elegans*, *Drosophila*, *Apis mellifera*) illustrates that behavioral equivalence does not imply representational equivalence. These notions generate three testable predictions linking training dynamics, representational geometry, and behavioral outcomes.

Our results suggest that grounding should be understood as a structural, measurable property, providing a bridge between cognitive theory and empirical analysis of artificial systems.

Keywords: symbol grounding, abduction, cognitive distance, negative abductive capability, representational similarity analysis, comparative cognition

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1 Introduction

Two questions about cognition have received sustained but largely separate treatment in the literature. The first is the *symbol grounding problem*: how do internal symbols acquire meaning anchored in sensorimotor experience [1]? The second is the *problem of abduction*: how do cognitive systems generate and revise hypotheses under uncertainty [2, 3]?

We propose a framework in which these questions are structurally connected: the topology of the sensory space determines both the geometry of cognitive representations and the dynamics of hypothesis revision.

The standard treatment of symbol grounding is binary: either a symbol system is causally connected to sensorimotor states (grounded) or it is not. We argue that this framing conceals the deeper question: grounded *in what*, and *how*? Two agents may each satisfy Harnad's condition while inhabiting mutually incommensurable representational spaces. The existence of grounding does not imply the equivalence of grounding.

The standard treatment of abduction presents hypothesis revision as correction: a bad hypothesis is detected and patched while the underlying representational architecture remains intact. We argue that this framing is structurally wrong: when an agent adopts a new hypothesis, that hypothesis restructures the interpretive frame through which all subsequent input is processed. Revision is therefore not correction but *overwrite*. A premature overwrite does not introduce a local error; it establishes a biased initial condition for all downstream cognition.

Connecting these two claims: the grounding-induced topology of a cognitive system is not fixed at birth but updated through abductive transitions. Premature overwrite is therefore a *dynamic source* of topological distortion—it increases d_{cog} not by mismatching sensory spaces but by committing to an interpretive frame before the evidence base is sufficient to determine the correct topology.

Reliable cognition requires both the right structure (*grounding equivalence*) and the right dynamics (*negative abductive capability*). A system that has the right sensory apparatus but commits prematurely will accumulate topological distortion over time. A system that suspends commitment appropriately preserves the structural integrity of its representational space.

Relation to prior work. Gärdenfors's theory of conceptual spaces [4] provides the closest precedent for a geometric approach to meaning. Our framework is complementary but distinct: we derive the geometry from sensory transduction rather than stipulating quality dimensions in advance, and we use topology rather than Euclidean convexity. The abduction literature from Peirce onward has framed hypothesis revision as search within a fixed representational space; our overwrite model replaces the fixed-space assumption.

Scope and falsifiability. This paper is a theoretical proposal, not a completed formal theory. The central claims are advanced as structural hypotheses supported by definitional argument and phenomenological evidence from comparative cognition; their full formal generalization is not established. We identify three classes of empirical evidence that would bear on the theory. First, the d_{cog} framework predicts that species sharing behavioral rules but not sensory topologies will show systematically higher d_{cog} as measured by representational similarity analysis [15]; this is testable via neural recording in behavioral paradigms that equate transfer accuracy across species. Second, the NAC hypothesis predicts that agents with higher commitment thresholds will show lower premature overwrite rates under ambiguous stimuli with gradually accumulating evidence; this is measurable as commitment latency and reversibility in both biological and artificial systems. Third, the overwrite-as-d_{cog}-increase hypothesis predicts that training conditions that suppress suspension (e.g., contrastive objectives) will produce higher d_{cog} relative to an ideally grounded reference, measurable via RSA on learned representations. Open Problems (Section 8) identify the

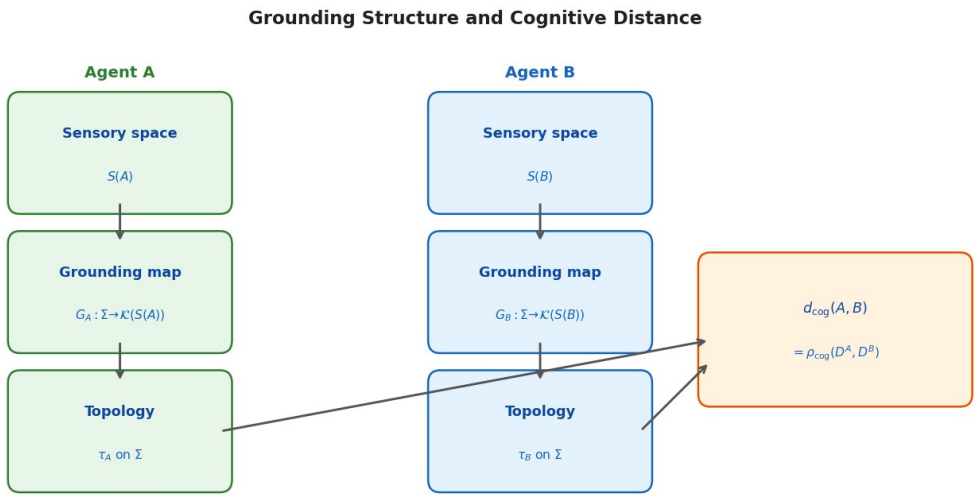

Figure 1: **Grounding structure and cognitive distance.** Each agent's sensory space $S(A)$ induces a topology τ_A on the shared symbol set Σ via the grounding map G_A. The cognitive distance $d_{\text{cog}}(A, B)$ measures the structural gap between τ_A and τ_B, operationalized as the second-order RSA correlation ρ_{cog} (see Remark 2).

most pressing formalization and measurement obligations.

Central variables. The two central variables of this paper are d_{cog} and NAC. d_{cog} is the static measure: how far apart are two agents' grounding-induced topologies? NAC is the dynamic condition: does the agent have the capacity to prevent that gap from widening through premature commitment? All other constructs in the paper—suspension region, commitment threshold, overwrite transition, bias propagation factor—are supporting definitions that make these two variables precise.

2 The Topology of Grounding

2.1 Sensory Spaces

Let A be a cognitive agent equipped with transduction channels $\mathcal{T} = \{T_1, \ldots, T_n\}$, where each T_i maps physical stimuli in domain Φ_i to internal activation states A_i.

Definition 1 (Sensory Space)**.** *The sensory space of agent A is the product space*

$$S(A) = \prod_{i=1}^{n} A_i$$

equipped with the product topology, where each A_i carries the topology induced by the neural activation metric of channel T_i.

Remark 1. *This definition is intentionally coarse. Cross-modal interactions and attentional modulation couple the channels; a refined treatment would replace the product with a fiber bundle. We use the product topology as a first approximation and flag the refinement as OP2.*

2.2 Grounding as a Map

Let Σ be a set of symbols. Let $\mathcal{K}(S(A))$ denote the hyperspace of nonempty closed subsets of $S(A)$, equipped with the Vietoris topology [5].

Definition 2 (Weak Grounding)**.** *A grounding map for A is a function $G : \Sigma \to \mathcal{K}(S(A))$ where $G(\sigma)$ is the closed set of sensory states associated with symbol σ. The symbol set Σ is* weakly grounded *in A if $G(\sigma) \neq \emptyset$ for all $\sigma \in \Sigma$.*

Weak grounding is Harnad's condition. It rules out purely formal symbol systems but says nothing about the structure of $G(\sigma)$ within $S(A)$.

2.3 Topological Equivalence of Grounding

Definition 3 (Grounding-Induced Topology)**.** *The grounding-induced topology τ_A on Σ is the initial topology with respect to G: the coarsest topology making $\sigma \mapsto G(\sigma)$ continuous with respect to the Vietoris topology on $\mathcal{K}(S(A))$.*

Definition 4 (Topological Equivalence of Grounding)**.** *Agents A and B are* groundingly equivalent *with respect to Σ if there exists a homeomorphism $h : (\Sigma, \tau_A) \to (\Sigma, \tau_B)$.*

Grounding equivalence is strictly stronger than weak grounding. Two agents may both be weakly grounded yet fail to be groundingly equivalent. This failure is what we call a *topological mismatch of grounding.*

2.4 Cognitive Distance

Conjecture 1 (Topological Sufficiency)**.** *If agents A and B are groundingly equivalent with respect to Σ, then they share every concept $\sigma \in \Sigma$ in the full intensional sense.*

Conjecture 2 (Topological Necessity)**.** *If agents A and B share a concept σ in the full intensional sense, then they are groundingly equivalent with respect to Σ.*

The contrapositive of Conjecture 2 is practically useful: if the sensory spaces of A and B are topologically inequivalent, then no bijection between their symbol sets constitutes a genuine translation.

In practice, cognitive systems share concepts to varying degrees. We therefore introduce a graded framework.

Definition 5 (Cognitive Distance; v2 signature form)**.** *The* cognitive distance $d_{\mathrm{cog}}(A, B)$ *is a measure of structural dissimilarity between the grounding-induced topologies τ_A and τ_B over a shared symbol set Σ. Following the persistence signature formulation of [**?**, Def. **??**], d_{cog} is the tuple*

$$d_{\mathrm{cog}}(A, B) = \Big(d_{\mathrm{GH}}\big((\Sigma, d_A), (\Sigma, d_B)\big),\ \{d_{\mathrm{bn}}^{(k)}(A, B)\}_{k\geq 0},\ \{\mathrm{PD}_k(A), \mathrm{PD}_k(B)\}_{k\geq 1} \Big),$$

combining Gromov–Hausdorff distance with bottleneck distances between k-th Vietoris–Rips persistence diagrams (see Appendix A for the full construction). In practice it is operationalized via the second-order RSA correlation ρ_{cog} defined in Remark 2, which serves as a tractable proxy for the $d_{\text{bn}}^{(0)}$ component. The three-level hierarchy of conceptual alignment is:

Remark 2 (RSA Operationalization and Second-Order Structural Similarity)**.** *Let $\Sigma = \{\sigma_1, \ldots, \sigma_n\}$ be a fixed probe concept set ($n = 29$ concepts spanning physical, abstract, relational, and temporal domains; see Appendix B). For agent A at hidden layer ℓ, define the RSA dissimilarity matrix*

$$D_{ij}^{A,\ell} = 1 - \text{corr}(\mathbf{h}_i^{A,\ell}, \mathbf{h}_j^{A,\ell}),$$

where $\mathbf{h}_i^{A,\ell} \in \mathbb{R}^d$ is the mean hidden-state activation at layer ℓ elicited by a canonical prompt for σ_i [15]. This defines a pseudo-metric space $(\Sigma, d^{A,\ell})$.

We adopt as primary operative metric the second-order RSA correlation

$$\rho_{\text{cog}}(A, B; \ell) = \text{Spearman}\big(\text{upper}(D^{A,\ell}), \text{upper}(D^{B,\ell})\big)\,,$$

which measures structural similarity independently of scale. Here $\text{upper}(\cdot)$ *denotes the upper-triangular entries. $\rho_{\text{cog}} = 1$ indicates identical relational geometry; $\rho_{\text{cog}} = 0$ indicates structural independence; $\rho_{\text{cog}} < 0$ indicates inverted relational structure.*

Pilot experiment (Pythia-70m). *We apply this operationalization to three publicly available training checkpoints of Pythia-70m [18]: step 0 (random initialization), step 1000 (early training), and step 143000 (fully trained reference). RSA matrices are computed at hidden layers $\ell \in \{2, 4, 6\}$ (early, middle, and final transformer layers) using $n = 29$ probe concepts. In all empirical analyses, d_{cog} is operationalized by ρ_{cog}; the Gromov–Hausdorff formulation serves as the theoretical idealization whose full construction appears in Appendix A. More precisely, under the signature form introduced above, ρ_{cog} is a monotone proxy for the $d_{\text{bn}}^{(0)}$ component of d_{cog} (merge-structure alignment); the higher-order components $d_{\text{bn}}^{(k)}$ ($k \geq 1$) require larger concept inventories than the twenty-nine–concept probe set to surface reliably (see [**?**, App. **??**]), and are outside the scope of the analyses below.*

Experiment 1: Training dynamics (Pythia-70m). *To validate Prediction P3, we compare three publicly available training checkpoints of Pythia-70m [18]: step 0 (random initialization), step 1000 (early training), and step 143000 (fully trained reference). RSA matrices are computed at hidden layers $\ell \in \{2, 4, 6\}$ using $n = 29$ probe concepts and three prompt templates ("The concept of {c} is", "A {c} is", "{c} refers to") averaged to reduce surface-form bias.*

Layer ℓ	$\rho_{\text{cog}}(\cdot, \text{ref})$		$\hat{d}_{\text{cog}}(\cdot, \text{ref})$	
	step-0	*step-1000*	*step-0*	*step-1000*
2	0.095	0.396	0.154	0.223
4	0.064	0.317	0.170	0.292
6	0.017	0.535	0.176	0.234

ρ_{cog} increases monotonically across all layers (step-0 < step-1000, confirmed at $\ell = 2, 4, 6$), consistent with Prediction P3. The near-zero variance of $D^{\text{ref},\ell}$ at $\ell = 6$ ($\sigma \approx 10^{-4}$) reflects the known representation isotropy of deeply-trained language models [17]—itself a measurable

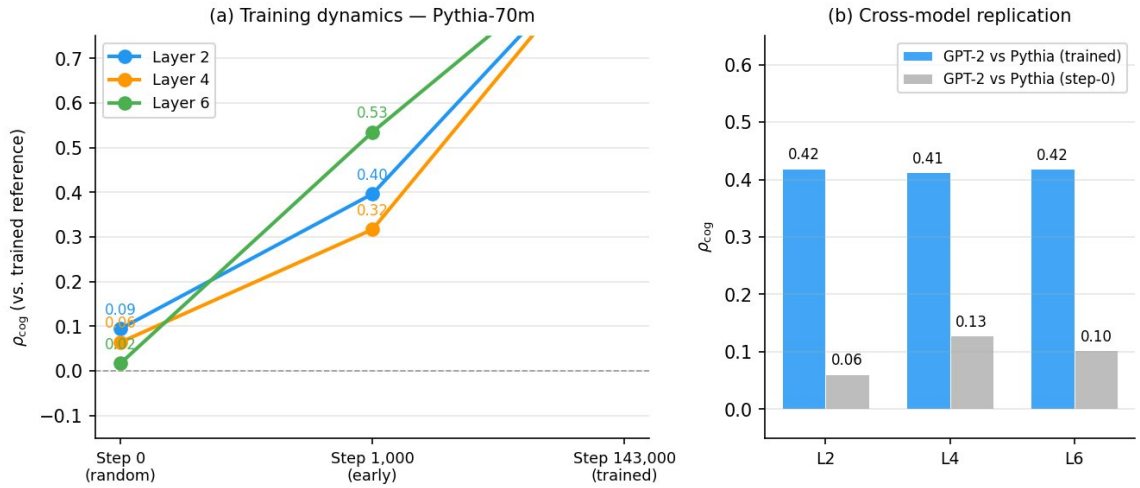

Figure 2: **RSA-based cognitive distance across training and models.** *(a)* ρ_{cog} between Pythia-70m checkpoints and the fully trained reference (step-143000), measured at layers 2, 4, and 6. All layers show monotonic increase, consistent with Prediction P3. *(b)* Cross-model comparison: ρ_{cog} between GPT-2 (trained) and Pythia at trained vs. random-initialization checkpoints. Trained models show substantially higher structural alignment ($\rho \approx 0.42$) than random initialization ($\rho \approx 0.10$), across all layers.

signature of the grounding-topology collapse described in Proposition 1—which renders $\hat{d}_{\text{cog}}$ *unreliable in this regime. We therefore adopt* ρ_{cog} *as the primary operative metric.*

Experiment 2: Cross-model replication (GPT-2). *To assess whether results are model-specific, we compute* ρ_{cog} *between GPT-2 small (trained) and Pythia-70m at both the trained and random-initialization checkpoints.*

Layer ℓ	*GPT-2 vs Pythia-trained*	*GPT-2 vs Pythia-step-0*
2	0.419	0.061
4	0.413	0.128
6	0.419	0.103

Across all layers, $\rho_{\text{cog}}(\textit{GPT-2}, \textit{Pythia-trained}) \gg \rho_{\text{cog}}(\textit{GPT-2}, \textit{Pythia-step-0})$ *(*≈ 0.42 *vs.* ≈ 0.10*, replicated at* $\ell = 2, 4, 6$*). Two architectures trained on distinct corpora converge to similar relational geometry over the probe concept set, while random initialization shares no such structure. This cross-model convergence is consistent with the claim that training induces a shared grounding topology, rather than an architecture-specific artifact.*

All code and data are available at `https://github.com/minamominamoto/grounding-without-reality`

Limitations. *The RSA-based operationalization captures relational similarity in activation space and serves as a proxy for grounding-induced topology. It does not fully characterize grounding structure: results are sensitive to prompt formulation, representation anisotropy, and model-specific embedding geometry. Accordingly,* ρ_{cog} *should be interpreted as an ordinal measure of structural similarity rather than a complete metric of grounding equivalence. Extension to biological systems, where activation patterns cannot be directly extracted, remains the primary empirical obligation of OP1.*

This yields a three-level hierarchy:

Level	Condition	Interpretation
1	$d_{\text{cog}} < \infty$, weak grounding	Extensional overlap only
2	d_{cog} small	Approximate concept sharing
3	$d_{\text{cog}} = 0$	Homeomorphism; full intensional sharing

3 Evidence from Comparative Cognition

The cases examined in this section are illustrative rather than confirmatory: they motivate the need for the framework by showing that the question of grounding structure arises in real biological systems, but they do not by themselves establish non-homeomorphism of grounding-induced topologies. That determination requires the empirical program of OP1. The cases are chosen because their sensory spaces are well characterized and their cognitive capacities have been studied experimentally, making them natural first test cases for the theoretical framework.

3.1 *Caenorhabditis elegans*: Minimal Grounding

C. elegans has exactly 302 neurons and approximately 7,000 synapses, fully mapped [8]. Its sensory apparatus consists primarily of chemosensory neurons. The organism exhibits chemotaxis—directed movement toward or away from chemical gradients—which can be formalized as a predict-and-correct loop: the animal generates an implicit prediction about gradient direction, detects deviation, and adjusts locomotion [9]. This constitutes the minimal implementation of an abductive-corrective loop.

We note a further implication: this loop does not merely implement abduction but *suspended abduction*. The organism does not commit to a directional hypothesis on the basis of a single reading; it maintains the hypothesis tentatively, continues to sample, and revises only when accumulated evidence warrants it. The 302-neuron architecture constitutes a candidate minimal biological implementation of negative abductive capability (Section 4): the predict-and-correct loop is structurally required to run multiple correction cycles before the organism commits to a locomotor decision, instantiating suspension in neural dynamics.

The sensory space $S(C.\ elegans)$ is low-dimensional, dominated by the chemosensory axis. The grounding-induced topology on whatever proto-symbol set the organism maintains is therefore a low-dimensional chemical topology.

Example 1. *The symbol* FOOD-SOURCE *in C. elegans is grounded in chemical gradients. In Homo sapiens, the same extensional referent is grounded in a high-dimensional space including visual shape, color, texture, weight, smell, taste, and social context. The grounding-induced topologies are plausibly non-equivalent; the "concept" is therefore unlikely to be shared in the intensional sense.*

3.2 *Drosophila melanogaster*: Intermediate Complexity

Drosophila possesses compound eyes sensitive to ultraviolet and polarized light, olfactory neurons, mechanoreceptors, and gustatory receptors. Recent connectome data [10] reveals

extensive cross-modal convergence zones in the central complex, where olfactory, visual, and proprioceptive information is combined nonlinearly. This suggests $S(\textit{Drosophila})$ has a non-trivial fiber bundle structure not captured by the product topology of Definition 1—a concrete empirical case where the fiber bundle refinement of OP2 becomes necessary.

3.3 *Apis mellifera*: Abstract Concepts in a Non-Human Sensory Space

The honeybee provides the most philosophically striking case. Bees learn abstract relational concepts—"same as" and "different from"—and transfer them to novel stimuli [11]. Bees understand zero as a quantity smaller than one [12] and can count up to four [13].

The bees' sensory space is dominated by olfactory transduction of volatile organic compounds, ultraviolet-sensitive photoreception, and mechanoreception of substrate vibration. The waggle dance encodes food-source location through bodily kinematics calibrated to flight energy—not abstract distance [14]—constituting weak grounding in the sense of Definition 2.

The bee's concept of SAME is grounded in a sensory space structured by olfactory, ultraviolet, and mechanoreceptive dimensions. The human concept of SAME is grounded in a space structured by visual, auditory, and tactile dimensions. Both agents possess the concept in an abstract sense, but the grounding-induced topologies plausibly differ. On the present framework, this supports the hypothesis that the intensional content is not shared—a conclusion that would follow if Conjecture 2 holds.

4 Abduction as Overwrite

4.1 Representational States and the Overwrite Transition

Let $\mathcal{R}$ denote the space of representational states of agent A. A representational state $R \in \mathcal{R}$ encodes the agent's current interpretation of sensory input: the active hypotheses, the assigned categories, the weighted priors over sensory signals.

Definition 6 (Interpretive Frame)**.** *The interpretive frame $F(R)$ induced by state R is the function $F(R) : \mathcal{X} \to \mathcal{Y}$ that maps sensory inputs $x \in \mathcal{X}$ to interpreted outputs $y \in \mathcal{Y}$ under the current representational state.*

Definition 7 (Categorical Distinctness)**.** *Two hypotheses h and h' are* categorially distinct *if the partitions of $\mathcal{X}$ they induce are topologically inequivalent: $\tau_{h'} \neq \tau_h$. A* correction *is a state change in which the induced topologies are homeomorphic; an* overwrite *is a state change in which they are not.*

Proposition 1 (Abductive Transitions are Overwrites)**.** *Let A adopt a new explanatory hypothesis h_{t+1} at time $t+1$ that supersedes h_t. If h_{t+1} is categorially distinct from h_t, then the resulting state transition is an overwrite, not a correction.*

4.2 Bias Propagation after Premature Overwrite

Definition 8 (Premature Overwrite)**.** *An overwrite transition $R_t \to R_{t+1}$ is premature if the evidence base E_t is insufficient to distinguish h_{t+1} from competing hypotheses at an acceptable confidence level θ: $P(h_{t+1} \mid E_t) < \theta$ while some $h^i \neq h_{t+1}$ satisfies $P(h^i \mid E_t) \geq \theta$.*

Proposition 2 (Bias Propagation)**.** *Let R_{t+1} result from a premature overwrite where h_{t+1} is categorially distinct from the ground-truth hypothesis h^*. Then for all subsequent inputs $x_{t'}$, $t' > t$, the interpreted output $F(R_{t+1})(x_{t'})$ carries systematic bias toward h_{t+1}.*

Conjecture 3 (Non-Reducibility of Categorical Bias)**.** *Within the committed frame $F(R_{t+1})$, the bias introduced by a categorially distinct premature overwrite is non-reducible by further Bayesian updating alone; it can only be eliminated by a subsequent overwrite transition. The toy model in Appendix C illustrates this claim in a two-hypothesis setting under fixed frame constraints. Full generality across hypothesis classes and frame structures constitutes an open problem (OP3).*

Remark 3 (Scope)**.** *The non-reducibility claim is restricted to categorially distinct hypotheses. When h_{t+1} and h^* induce homeomorphic partitions of $\mathcal{X}$, standard Bayesian updating will converge to the correct posterior. The present claim applies specifically to cases in which the adopted hypothesis restructures the topology of the interpretive frame.*

Corollary 1. *In the categorially distinct case, a premature overwrite is not recoverable by Bayesian updating within the committed frame. It constitutes a structural initial condition that propagates interpretive bias until a subsequent overwrite occurs. This corollary is conditional on the non-reducibility conjecture in Proposition 2; its empirical status is assessed via Predictions P2 and P3.*

4.3 Premature Overwrite as a Dynamic Source of d_{cog} Increase

The grounding-induced topology τ_A is not fixed; it is updated by abductive transitions. To make this precise, we note the following structural connection between overwrite transitions and changes in τ_A.

Proposition 3 (Overwrite May Induce Topology Change)**.** *Let A undergo an overwrite transition from state R_t to R_{t+1}, where the adopted hypothesis h_{t+1} is categorially distinct from h_t (Definition 7). Then the grounding map updates: $G_{t+1}(\sigma) \neq G_t(\sigma)$ for at least one $\sigma \in \Sigma$. Under the assumption that this update induces a non-homeomorphic reorganization of symbol neighborhoods—that is, that the updated grounding map does not happen to preserve the initial topology—the grounding-induced topology changes: $\tau_{t+1} \neq \tau_t$, and the two states imply positive cognitive distance: $d_{\text{cog}}(A_t, A_{t+1}) > 0$ under this assumption.*

Remark 4. *The conditional in Proposition 3 is non-trivial: it is conceivable that a categorially distinct overwrite changes the grounding map while leaving the induced initial topology invariant, if the new and old maps happen to generate the same open sets on Σ. We regard this as a degenerate case unlikely to arise in systems with rich symbol sets, but acknowledge it as a formal gap. The assumption is satisfied in the Appendix A toy model, where the two*

hypotheses induce genuinely distinct partitions of $\mathbb{R}^2$ with non-homeomorphic induced topologies. The general condition under which a grounding map update is guaranteed to change τ_A is listed as OP6.

Premature overwrite therefore constitutes a *dynamic* source of d_{cog} increase: the topology of the committed frame deviates from the topology of the ideal frame, and this deviation propagates until the next overwrite.

This connects the two parts of the paper. Topological mismatch between agents (Section 2) has two sources: (i) structural mismatch in sensory space, and (ii) differential abductive history—agents that commit at different rates or under different evidence conditions will accumulate different topological distortions over time. d_{cog} is therefore a dynamic quantity, not merely a static comparison of sensory spaces.

5 Negative Abductive Capability

5.1 Definition

Definition 9 (Suspension Region)**.** *The suspension region $\mathcal{S} \subset \mathcal{R}$ is the set of representational states in which the agent maintains active hypotheses tentatively without committing to any single overwrite. An agent in a suspension state accumulates evidence and updates probability weights without triggering a frame-restructuring transition.*

Definition 10 (Commitment Threshold)**.** *The commitment threshold $\theta^* \in (0, 1)$ is the minimum posterior probability required to trigger an overwrite transition.*

Definition 11 (Negative Abductive Capability (NAC))**.** *An agent A possesses negative abductive capability with respect to hypothesis class $\mathcal{H}$ if:*

1. *A maintains a well-defined suspension region $\mathcal{S}$;*
2. *A's commitment threshold θ^* is calibrated to the evidence requirements of $\mathcal{H}$; and*
3. *A can remain in $\mathcal{S}$ for an extended observation window without degradation of suspension capacity.*

The term is chosen by analogy with Keats's negative capability— the ability to remain "in uncertainties, mysteries, doubts, without any irritable reaching after fact and reason" [7]— recast here as a structural cognitive property rather than a poetic temperament.

Remark 5. *NAC is not the absence of abductive capacity. An agent with high NAC can generate hypotheses, hold them in suspension, and commit only when evidence warrants. An agent with low NAC commits rapidly, accumulating bias.*

Proposition 4 (NAC as Structural Prerequisite)**.** *Under the overwrite model, the expected bias accumulated per abductive cycle is a decreasing function of NAC and is conjectured to be independent of hypothesis-generation capacity. This structural hypothesis implies that abductive reliability cannot be improved by increasing generation capacity alone; NAC must be co-present.*

Central thesis. *Abduction requires negative capability. The power to infer depends on the power to suspend inference.*

5.2 NAC and d_{cog}

An agent with NAC calibrated to $\mathcal{H}$ remains in suspension until $t \geq t^{\dagger}$ (the time at which evidence warrants commitment). The committed frame $F(R_{t^{\dagger}})$ is then homeomorphic to the ideal frame $F(R^*)$, and $d_{\text{cog}}(A, A^*) = 0$. NAC is therefore a sufficient structural condition for d_{cog} minimization within the present model.

6 Empirical Predictions

The framework yields the following predictions, each connecting a theoretical construct to a measurable quantity.

P1. Behavioral equivalence does not imply representational equivalence. Two species that demonstrate matched behavioral transfer accuracy on an abstract relational rule (SAME, DIFFERENT, ZERO) may nevertheless differ in d_{cog}. *Measurement*: Collect neural activation patterns in each species during rule application; compute representational dissimilarity matrices via RSA [15]; compare the resulting metric spaces via Gromov–Hausdorff distance. *Prediction*: d_{cog} will be positive and inversely correlated with sensory space overlap across species pairs matched for behavioral accuracy.

P2. NAC suppression increases premature commitment rate. Agents or systems whose suspension capacity is structurally suppressed (high PCR) will commit to interpretive frames earlier under ambiguous or gradually accumulating evidence than agents with high NAC. *Measurement (biological)*: Present ambiguous stimuli (e.g., multistable perceptual displays; contradictory chemical gradients for *C. elegans*) with controlled evidence accumulation rate; measure commitment latency (time to stable behavioral choice) and reversibility (degree to which contradictory evidence induces re-suspension). *Measurement (AI)*: Use attribution graphs from mechanistic interpretability tools [16] to identify whether multiple competing hypotheses are simultaneously active (suspension) or one hypothesis suppresses others (commitment); measure the transition latency as a function of evidence quality. *Prediction*: Systems trained with contrastive objectives will show shorter commitment latency and lower reversibility than systems trained with calibrated uncertainty objectives.

P3. Training topology determines d_{cog}, not only accuracy. Two systems trained to the same behavioral accuracy on a grounding task but under different training topologies (different sequences or distributions of sensory exposures) will differ in d_{cog} relative to a reference grounding. *Measurement*: Train two models on matched accuracy targets but with early vs. late presentation of informative stimuli (manipulating premature overwrite rate); measure d_{cog} via RSA against a fully-informed reference model. *Prediction*: Models exposed to early informative stimuli (high premature overwrite risk) will show higher d_{cog} despite matched accuracy.

7 Implications

7.1 Cross-Species Concept Attribution

Behavioral experiments establish that an organism has learned a rule (SAME, DIFFERENT, ZERO) under specific training conditions. The grounding map G is formed through this training history. If training conditions are informationally impoverished relative to the full structure of the concept, the grounding-induced topology τ_A will reflect the training experience rather than the full intensional structure.

Two consequences follow. First, d_{cog} measured in behavioral experiments is a lower bound on the true cognitive distance. Second, the grounding maps may carry the signature of premature abduction—early hypotheses that bias all further sensory interpretation.

Cross-species concept commensurability studies should report not only behavioral transfer accuracy but also the *training topology*: the sequence and informativeness of sensory exposures through which the concept was formed.

7.2 Refining the Critique of Multimodal AI Systems

The framework provides a more precise diagnostic for AI systems than the standard "grounded vs. ungrounded" binary. This is not primarily an argument about AI, but a test case for the theory: if the framework is correct, AI systems that lack NAC should show predictable patterns of d_{cog} elevation relative to biological reference groundings.

Adding sensory channels to an AI system does not automatically produce a sensory space topologically equivalent to the human sensory space. Moreover, current multimodal systems commit to grounding assignments via weight updates on each training batch, regardless of whether the evidence in that batch is sufficient to determine the correct interpretive frame. The result is premature overwrite at scale: the model's τ_A reflects training distribution artifacts rather than sensory structure.

Current systems also lack a well-defined suspension region. We conjecture that contrastive training objectives (e.g., CLIP-style alignment) suppress NAC by rewarding confident cross-modal matching rather than calibrated uncertainty; this suppression of suspension capacity is itself a prediction of the framework (P2) and should be tested empirically rather than assumed.

This requires distinguishing two superficially similar states that are easy to conflate in current AI systems:

- **Uncertainty**: the agent's posterior over hypotheses is wide—multiple hypotheses have non-negligible probability. This is a common state in transformer models and does not by itself constitute suspension.

- **Suspension**: the agent maintains competing hypotheses as active candidates *without triggering frame selection*— the competition is reversible and is maintained across further evidence accumulation rather than resolved by the first confident signal. The operational sign is reversible competition maintained under accumulating evidence, not transient co-activation.

Uncertainty is necessary but not sufficient for suspension in the formal sense of Definition 11. A system that shows wide posteriors but immediately commits as soon as any signal tips the balance has low NAC, even if its uncertainty is calibrated. This distinction determines what attribution graph analysis (P2) would actually need to find: not co-activation of competing features per se, but sustained, reversible co-activation that persists under continued evidence accumulation.

This is a falsifiable prediction: systems with genuine suspension capacity should show measurably higher commitment latency and reversal rate under ambiguous stimuli than systems without it.

7.3 Trustworthy AI: Beyond Accuracy

A trustworthy AI system is not merely one with high accuracy. It is one that does not prematurely fix its interpretive frame under insufficient evidence. We propose the following additional evaluation metrics:

1. **Premature commitment rate (PCR)**: fraction of overwrite transitions triggered below threshold θ^*.
2. **Overwrite reversibility (OR)**: extent to which a prior overwrite can be undone by subsequent evidence.
3. **Uncertainty retention (UR)**: duration over which the system maintains a suspension state without degradation.
4. **Bias propagation factor (BPF)**: amplification of interpretive bias across successive overwrite cycles.

8 Open Problems

The framework developed here is a theoretical proposal rather than a completed formal theory. The claims are advanced as structural hypotheses supported by definitional argument and phenomenological evidence from comparative cognition. Their full generality across cognitive architectures, hypothesis classes, and sensory modalities is not established and constitutes the primary research obligation of the program.

OP1. Measurement of grounding-induced topology. How can τ_A be empirically measured for a biological or artificial system? Candidate approach: representational similarity analysis [15] to construct a metric on Σ from neural activation patterns, then compare via Gromov–Hausdorff distance.

OP2. Fiber bundle refinement. Definition 1 uses the product topology, ignoring cross-modal coupling. A fiber bundle formulation, where the base space captures the dominant modality and fibers capture cross-modal modulation, is more biologically faithful. The *Drosophila* central complex [10] provides a concrete test case.

OP3. Proof of Conjecture 2 (Topological Necessity). A counterexample would require two agents with topologically inequivalent grounding yet identical relational structure over Σ.

OP4. Minimum neuron count for suspended abduction. The predict-and-correct loop in *C. elegans* runs in 302 neurons. What is the minimum network size to implement a suspension region of depth m for a hypothesis class of size k?

OP5. NAC measurement. For biological agents: measure commitment latency and reversibility under ambiguous stimuli with varying time pressure. For AI systems: Anthropic's circuit tracing tools [16] may reveal whether internal states corresponding to maintained hypothesis competition (suspension) exist in trained models.

OP6. Formal relationship between NAC and d_{cog}. Establish a quantitative relationship between the premature commitment rate of a system and the resulting increase in d_{cog} relative to an ideally grounded reference.

9 Conclusion

Symbol grounding is not a binary property. Whether a symbol system is grounded tells us only that some connection exists between symbols and sensorimotor states—it tells us nothing about the structure of that connection. And the structure of that connection is not fixed: it is updated through abductive transitions that may preserve or distort the grounding-induced topology depending on when commitment is made.

We have introduced two formal tools for making this precise. The cognitive distance d_{cog} measures the structural gap between grounding-induced topologies and provides a continuous measure of conceptual commensurability. Negative abductive capability (NAC) characterizes the structural capacity required to prevent premature topological distortion through abductive overwrite.

Together, these tools yield a structural account of reliable cognition: a system reliably shares concepts with others if its grounding-induced topology matches theirs (d_{cog} small), and it maintains that match over time by committing to interpretive frames only when the evidence base warrants it (NAC sufficient, within the present model).

The symbol grounding problem, thirty-six years after its formulation, may need to become the *symbol grounding structure problem*— and the abduction problem may need to become the *abductive dynamics problem.*

Abduction requires negative capability. The power to infer depends on the power to suspend inference.

Acknowledgments

No funding was received for this work.

A Formal Construction of Cognitive Distance via Gromov–Hausdorff Theory

The formal definition of d_{cog} proceeds as follows. Let $S(A) = \prod_{i=1}^{k} T_i$ be the sensory space of agent A, equipped with the product topology from transduction channels T_i. The grounding map $G : \Sigma \to \mathcal{K}(S(A))$ assigns each symbol $\sigma \in \Sigma$ a compact subset of sensory states, endowed with the Vietoris topology. This induces a topology τ_A on Σ: the coarsest topology making each $G(\sigma)$ continuous with respect to the Hausdorff metric on $\mathcal{K}(S(A))$.

To compare two agents A and B, one constructs compatible metrics d^A and d^B on Σ from τ_A and τ_B respectively (OP1 in Appendix A), then defines:

$$d_{\text{cog}}(A, B) = d_{GH}((\Sigma, d^A), (\Sigma, d^B)),$$

where d_{GH} is the Gromov–Hausdorff distance [6]. This yields $d_{\text{cog}} = 0$ if and only if (Σ, d^A) and (Σ, d^B) are isometric—corresponding to full intensional sharing of all concepts in Σ.

Computing exact Gromov–Hausdorff distances is NP-hard in general [19]. The RSA-based operationalization (Remark 2) provides a tractable proxy: under the assumption that the Pearson correlation structure of hidden-state activations approximates the relational geometry of τ_A, the second-order RSA correlation ρ_{cog} serves as an ordinal surrogate for d_{GH}. The Frobenius surrogate $\hat{d}_{\text{cog}} = \|D^A - D^B\|_F/n$ is a computable upper bound on d_{GH} when activation variances are comparable across agents, but degrades when one matrix collapses (as observed at the final layer of deeply-trained models; see Remark 2).

B Probe Concept Set

The following $n = 29$ concepts constitute the fixed probe set Σ used in the RSA operationalization (Remark 2). Concepts were selected to span four semantic domains, following the cross-domain coverage principle of Kriegeskorte et al. [15].

Domain	Concepts
Physical / perceptual	cat, dog, apple, fire, water, stone, sky, tree
Abstract	justice, truth, freedom, beauty, number, cause
Relational	same, different, larger, smaller, before, after
Emotional / social	fear, trust, anger, love
Cognitive	belief, doubt, memory, knowledge, inference

For each concept σ_i, activations are elicited using the prompt template "`The concept of {concept} is`", and the mean last-layer hidden state over all token positions is recorded as $\mathbf{h}_i^A$. All implementation details and code are available at `https://github.com/minamominamoto/grounding-without-reality`.

C Toy Model: Bayesian Overwrite and d_{cog}

We provide a minimal formal instantiation of Propositions 1 and 2 in a two-hypothesis Bayesian setting.

Setup. Let $\mathcal{H} = \{h_0, h_1\}$. Let $\mathcal{X} = \mathbb{R}^2$ with two regions $\mathcal{X}_0$ and $\mathcal{X}_1$, associated with h_0 and h_1 respectively. Observations $x_1, x_2, \ldots$ are drawn from a mixture distribution.

Suspension and commitment. The agent performs Bayesian updates while $\max_i P(h_i \mid x_1, \ldots, x_t) < \theta^*$. At time t^*, the first time a hypothesis exceeds θ^*, the agent commits: the interpretive frame is set to $p(\cdot \mid h_j)$.

Premature overwrite. If $t^* < t^\dagger$ (the time at which the maximum-likelihood hypothesis becomes stable under the full evidence sequence), the overwrite is premature.

Bias propagation. After commitment to h_j at t^*, the bias for subsequent input $x_{t'}$ is:

$$\beta(x_{t'}) = F(R_{t^*})(x_{t'}) - F(R^*)(x_{t'}).$$

For inputs from the non-committed region $\mathcal{X}_{1-j}$, $\beta(x_{t'}) \neq 0$ and is not correctable by Bayesian update within the committed frame.

Connection to d_{cog}. When $h_j \neq h^\dagger$ (premature commitment to the wrong hypothesis), the committed and ideal topologies are not homeomorphic:

$$d_{\text{cog}}(A, A^*) = d_{GH}((\Sigma, \tau_A), (\Sigma, \tau^*)) > 0.$$

An agent with NAC remains in suspension until $t \geq t^\dagger$, at which point $d_{\text{cog}}(A, A^*) = 0$.

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